

Mohamed Ali Saber

Mechatronics Engineering — Industrial Automation & IoT

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Professional Summary

Mechatronics Engineer specializing in Industrial Automation, Control Systems, and IoT-based smart solutions. Hands-on experience with Siemens PLCs (TIA Portal), Ladder Logic programming, PID control, and real-time embedded systems.

Built multiple real-time IoT systems using ESP32 and MQTT communication, integrating hardware sensors with cloud-based monitoring dashboards.

Winner of multiple IEEE robotics competitions (1st Place IoT, 1st Place Fire Fighting, 2nd Place Micromouse), demonstrating strong capabilities in system integration, control logic design, and industrial problem solving.

Education

Bachelor of Mechatronics Engineering — Mansoura University

GPA: 3.66 / 4.00

Industrial Automation & Control Systems

- Siemens S7-1200 PLC Programming using TIA Portal
- Ladder Logic (LAD), PLC Diagnostics and Troubleshooting
- Control Circuit Design using Relays and Contactors
- Motor Control Systems (Star-Delta, Forward/Reverse, VFD, Servo, Stepper)
- Industrial Timers (ON-Delay, OFF-Delay, Cyclic Timers)
- Industrial Sensors (Proximity, Encoders, Limit Switches)
- PID Control and Closed-loop Systems
- Factory I/O Simulation (Industrial Process Simulation)

IoT & Smart Systems

- ESP32 / STM32 Embedded Systems
- MQTT Protocol for Real-Time Data Communication
- Sensor Integration (IMU, GPS, Analog & Digital Sensors)
- Cloud-based Monitoring Dashboards
- Real-Time Data Processing using .NET Core Backend & Firebase

Projects

- **Industrial Conveyor Automation (Factory I/O)** Designed and implemented an automated industrial sorting line using Siemens S7 PLC in TIA Portal. Developed Ladder Logic for object detection, sorting sequences, conveyor synchronization, VFD motor speed control.
- **Graduation Project — CrashXpert (IoT Smart System)** Developed a real-time accident detection and response system using ESP32 with accelerometer and GPS sensors. Implemented

MQTT-based communication and built backend services using .NET Core for real-time monitoring and alerting.

- **Micromouse Autonomous Robot** Implemented real-time PID control, encoder feedback systems, and flood-fill navigation algorithm. Designed efficient motor control and sensor logic for autonomous navigation.

Experience

Head of Software Circle — **Momentum MU** **Sept 2023 – Sept 2024**

- Managed integration between embedded systems, control algorithms, and mechanical subsystems.

Software Instructor — **Luminous MU**

- Mentored students in embedded systems, robotics programming, and software engineering fundamentals.

Technical Skills

- **Programming:** C, C#, Python
- **Automation Tools:** TIA Portal, Factory I/O
- **Embedded:** STM32, ESP32, Arduino
- **Communication:** UART, SPI, I2C, MQTT
- **Software:** .NET Core, Git, GitHub

Key Achievements & Competitions

- 1st Place — IEEE IoT Competition
- 1st Place — IEEE Fire Fighting Robotics Competition
- 2nd Place — IEEE Micromouse Competition (Best Software Performance)

Trainings

- Full Stack .NET Diploma — DEPI (6 Months)
- Machine Learning & Deep Learning Diploma — NTI
- Classic Control Course
- Embedded Systems Course
- Siemens PLC Course

Languages

- Arabic — Native
- English — Intermediate